Arduino Tx/Rx + 433MHz RF Module

RF BASED SMART AGRICULTURE MONITORING SYSTEM

Project Team & Academic Details

Team Members

Student 1: -Aryan Singh Sikarwar
Student 2:- purav patel
Student 3:- Ajay Kumar yadav
Project Guide:- Rajalakshmi S

Institution Details

Department of Electronics & Communication Engineering
[VIT VELLORE], [VELLORE] | Academic Year 2025-26

Aim & Objectives

Project Aim

To develop an offline wireless smart agriculture monitoring system that enables real-time farm data collection and transmission without internet dependency.

Key Objectives

Implement RF-based communication between field sensors and home monitoring unit for seamless wireless data transfer using 433MHz modules.

Literature Survey

& Research Background

IoT in Agriculture

Smart farming leverages sensor networks for real-time crop monitoring, reducing water waste by 30-50% through precision irrigation.

RF Communication Systems

433MHz RF modules enable low-cost, license-free wireless data transmission up to 500m, ideal for offline agricultural environments.

Sensor-Based Monitoring

Multi-sensor integration (soil moisture, temperature, gas detection) provides comprehensive environmental data for automated farm management.

Websites Referred & References

Arduino Official Documentation

arduino.cc - Comprehensive guides for Arduino Uno programming, sensor interfacing, and serial communication protocols.

Sensor Datasheets & Specs

DHT-22, YL-69, MQ-5, MQ-135 technical datasheets from component manufacturers for calibration and integration.

RF Module Resources

433MHz RF Tx/Rx module specifications, VirtualWire library documentation, and IEEE research papers on wireless sensor networks.

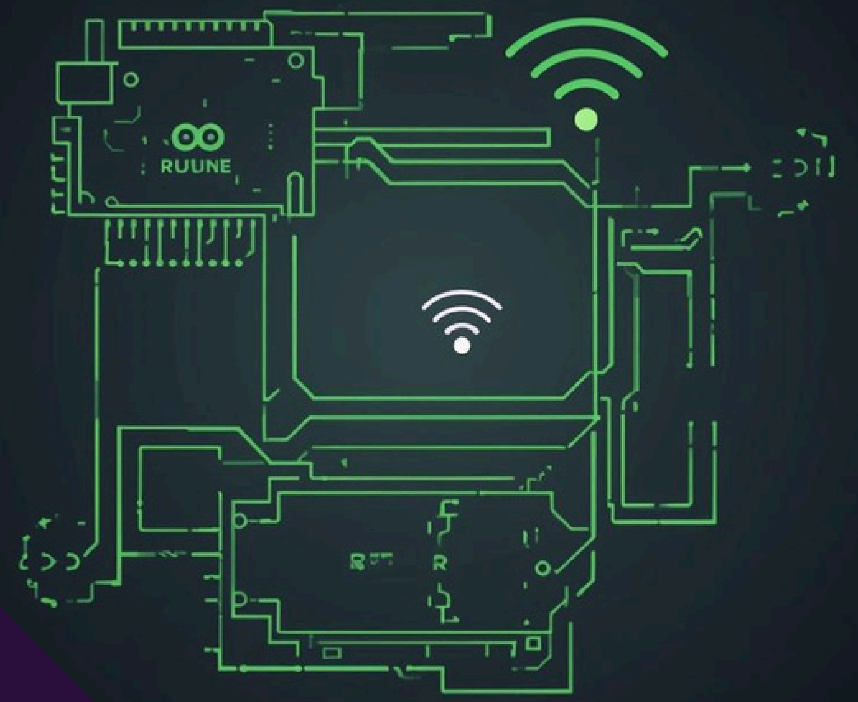
System Architecture
& Design Overview

Field Unit (Transmitter)

Arduino Uno reads sensor data from YL-69, DHT-22, MQ-5, and MQ-135 modules, then transmits via 433MHz RF module.

Home Unit (Receiver)

Arduino Uno receives RF signals, displays live data on Serial Monitor, and triggers buzzer alerts for critical conditions.



System Block Diagram & Data Flow

Phase 1: Sensor Data Acquisition

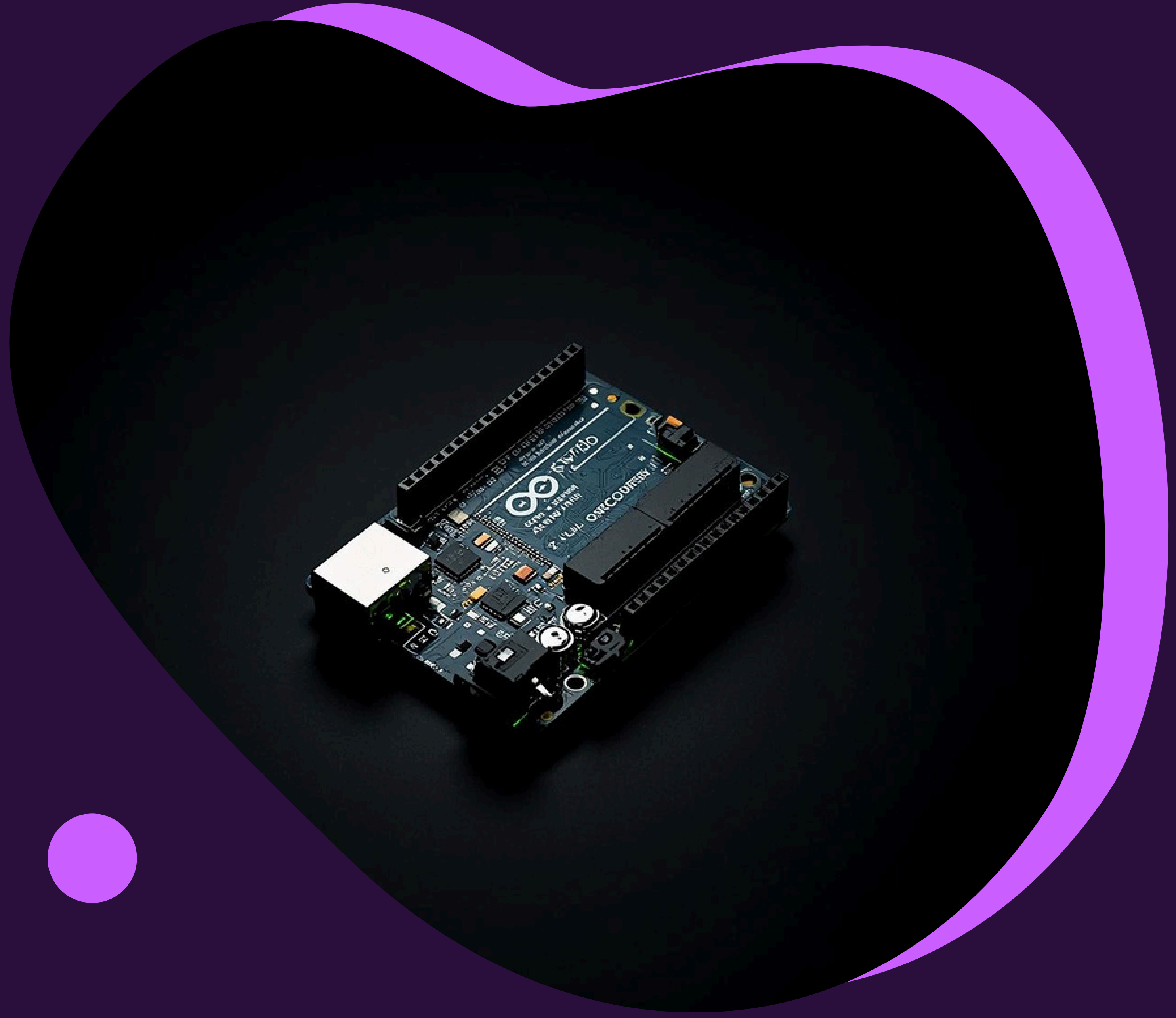
Field Unit Arduino reads real-time data from YL-69 soil moisture, DHT-22 temp/humidity, MQ-5 gas, and MQ-135 air quality sensors.

Phase 2: RF Wireless Transmission

Sensor readings are encoded and transmitted wirelessly via 433MHz RF transmitter module to the Home Unit receiver.

Phase 3: Reception & Alert System

Home Unit Arduino receives RF data, displays live readings on Serial Monitor, and triggers buzzer alerts for critical thresholds.



Field Unit Hardware

Transmitter Components

Arduino Uno Microcontroller

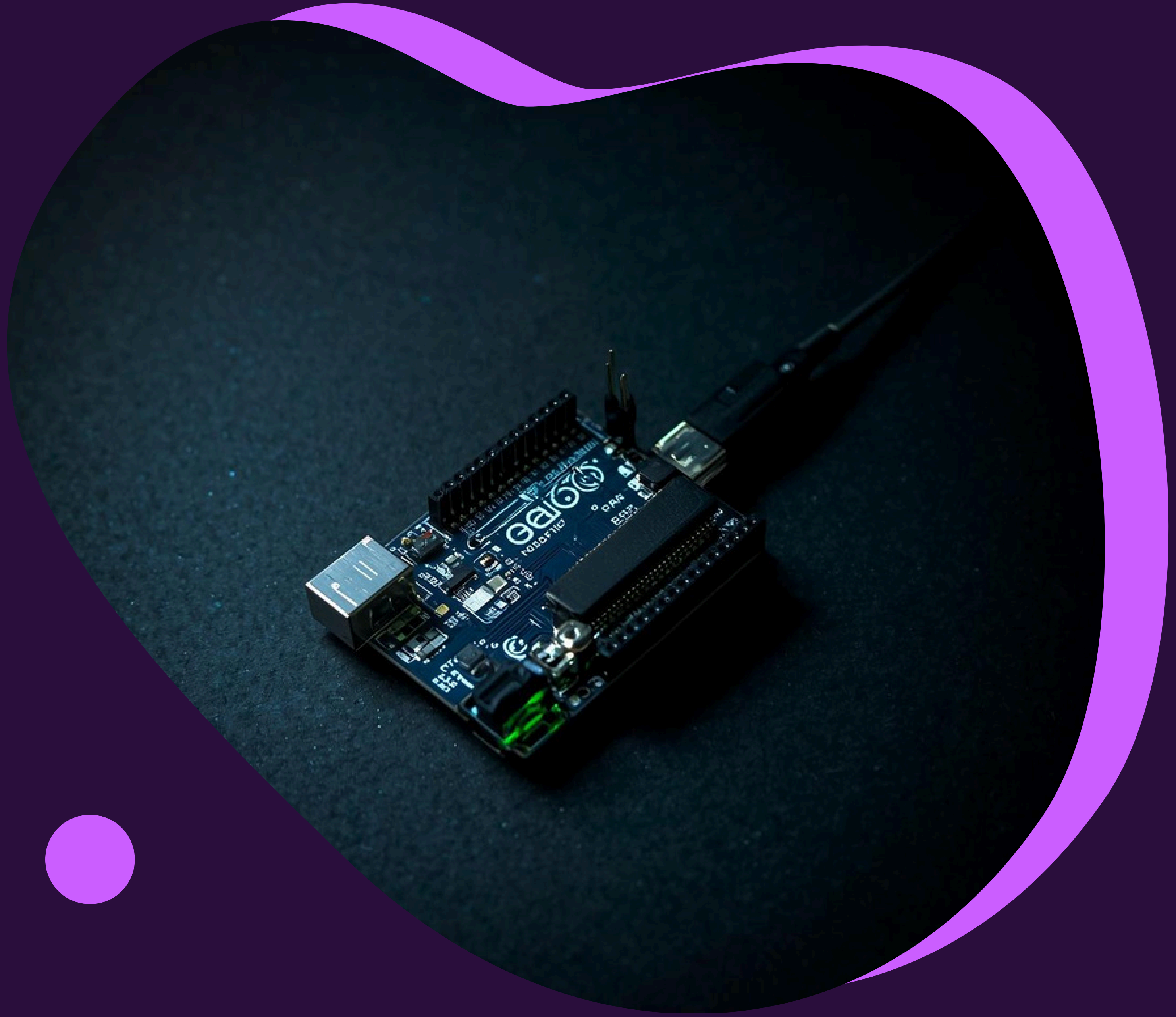
Central processing unit that reads all sensor data and controls RF transmission to the home unit.

YL-69 Soil Moisture Sensor

Measures soil moisture levels to determine irrigation needs and triggers alerts when soil is too dry.

DHT-22 Temperature & Humidity

High-precision sensor providing accurate ambient temperature and humidity readings for crop monitoring.



Hardware Components
Part 2: Home Unit

Arduino Uno (Receiver)

Microcontroller brain for Home Unit; receives RF data, processes sensor alerts, and outputs to Serial Monitor.

433MHz RF Receiver Module

Wireless receiver paired with field transmitter; decodes incoming sensor data packets at 433MHz frequency.

MQ-5 & MQ-135 + Buzzer

Gas/fire detection (MQ-5) and air quality monitoring (MQ-135) sensors; buzzer provides audible alerts for critical thresholds.

Software Details

Arduino IDE Platform

Primary development environment for programming both Tx/Rx Arduino Uno boards with C/C++ code for sensor reading and RF communication.

Serial Monitor & Alerts

Live data visualization via Serial Monitor at 9600 baud; conditional logic triggers buzzer alerts when soil moisture drops or gas levels exceed thresholds.

System Algorithm & Flowchart

Phase 1: Sensor Initialization

Arduino powers up and initializes all sensors: YL-69 (soil), DHT-22 (temp/humidity), MQ-5 (gas), MQ-135 (air quality).

Phase 2: Data Read & RF Transmission

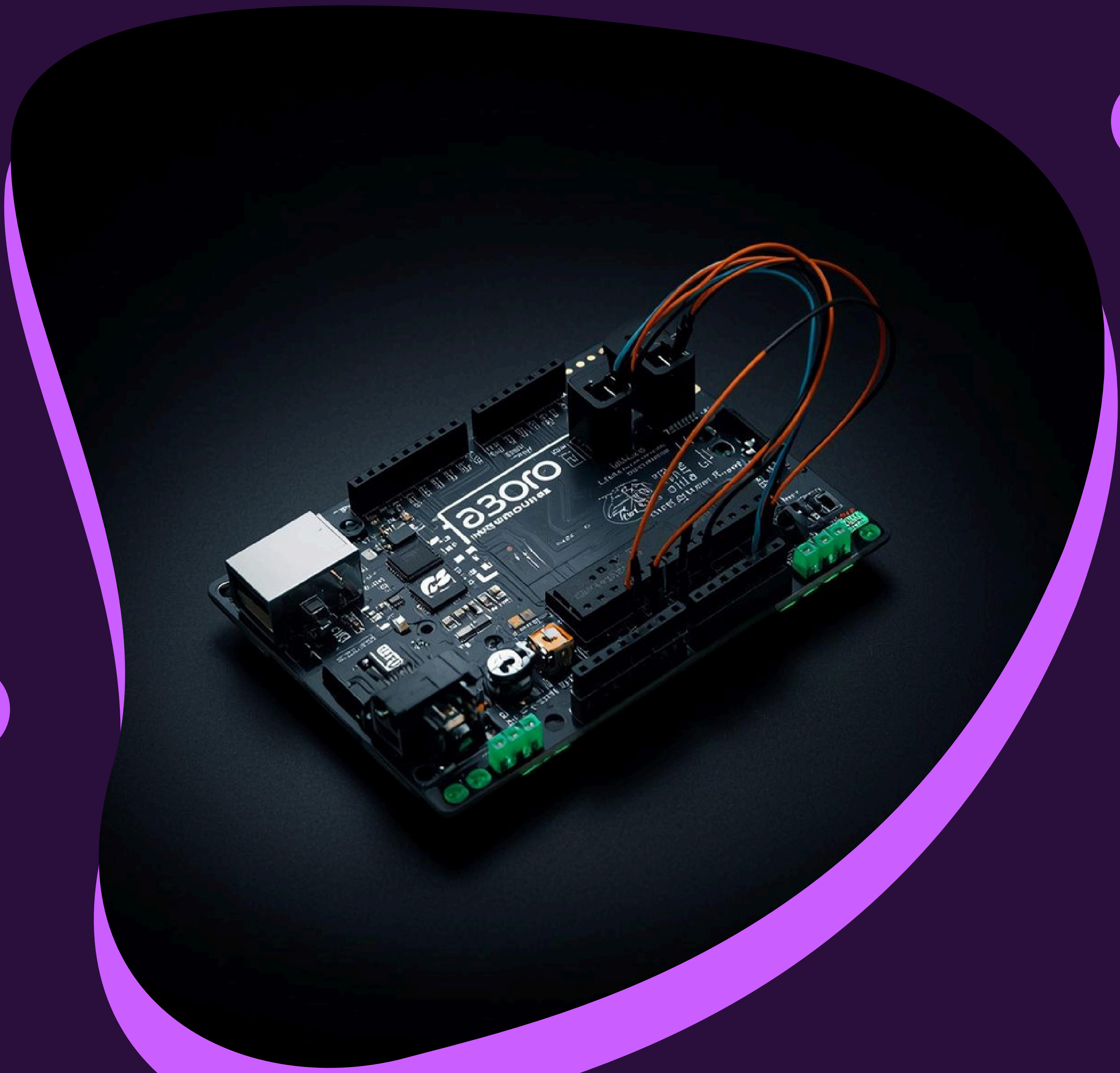
Field Unit reads sensor values in loop, packages data, and transmits wirelessly via 433MHz RF module to Home Unit.

Phase 3: Receive, Process & Alert

Home Unit receives RF data, displays on Serial Monitor, analyzes thresholds, and triggers buzzer if soil dry or gas detected.

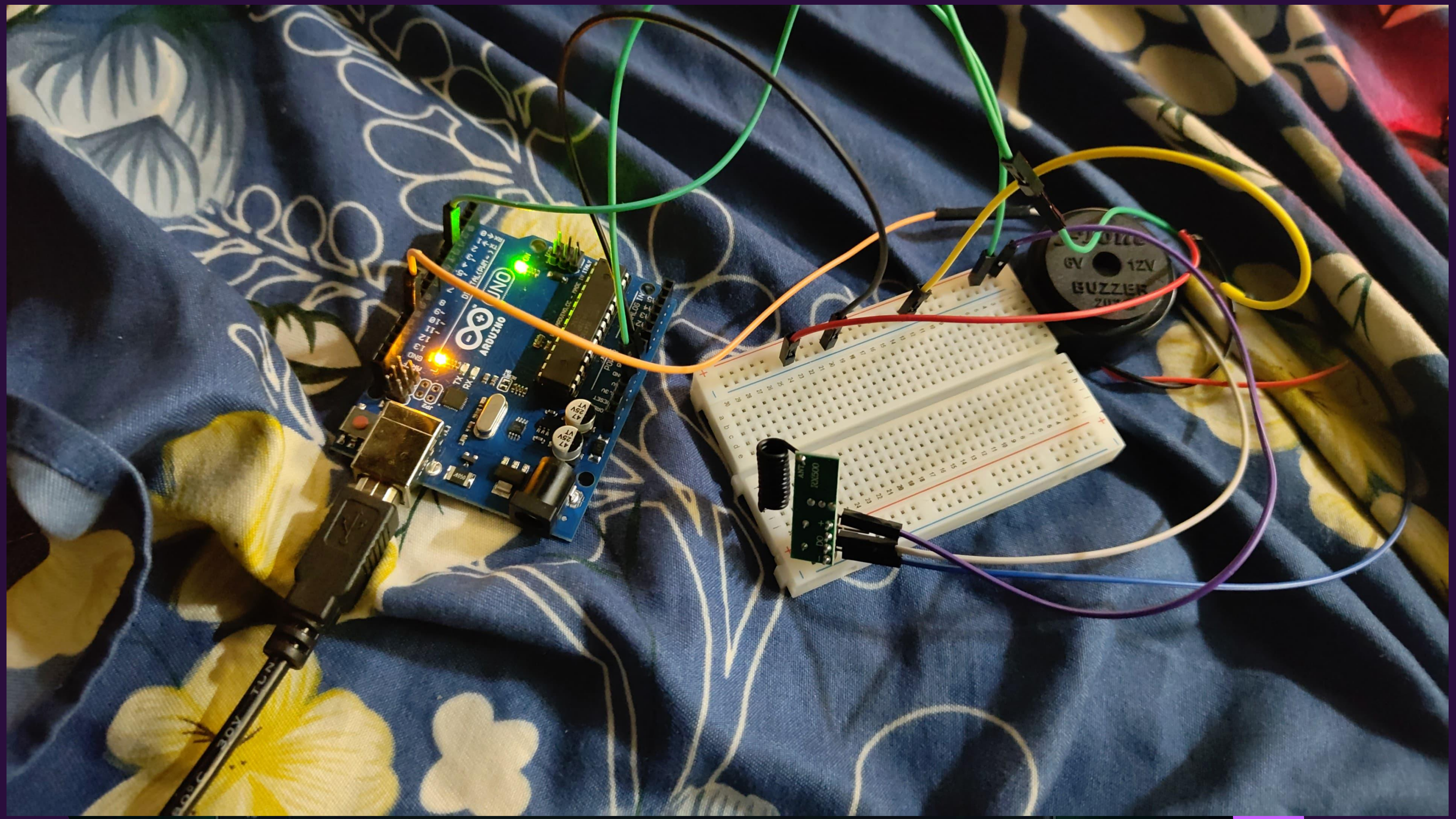
Work

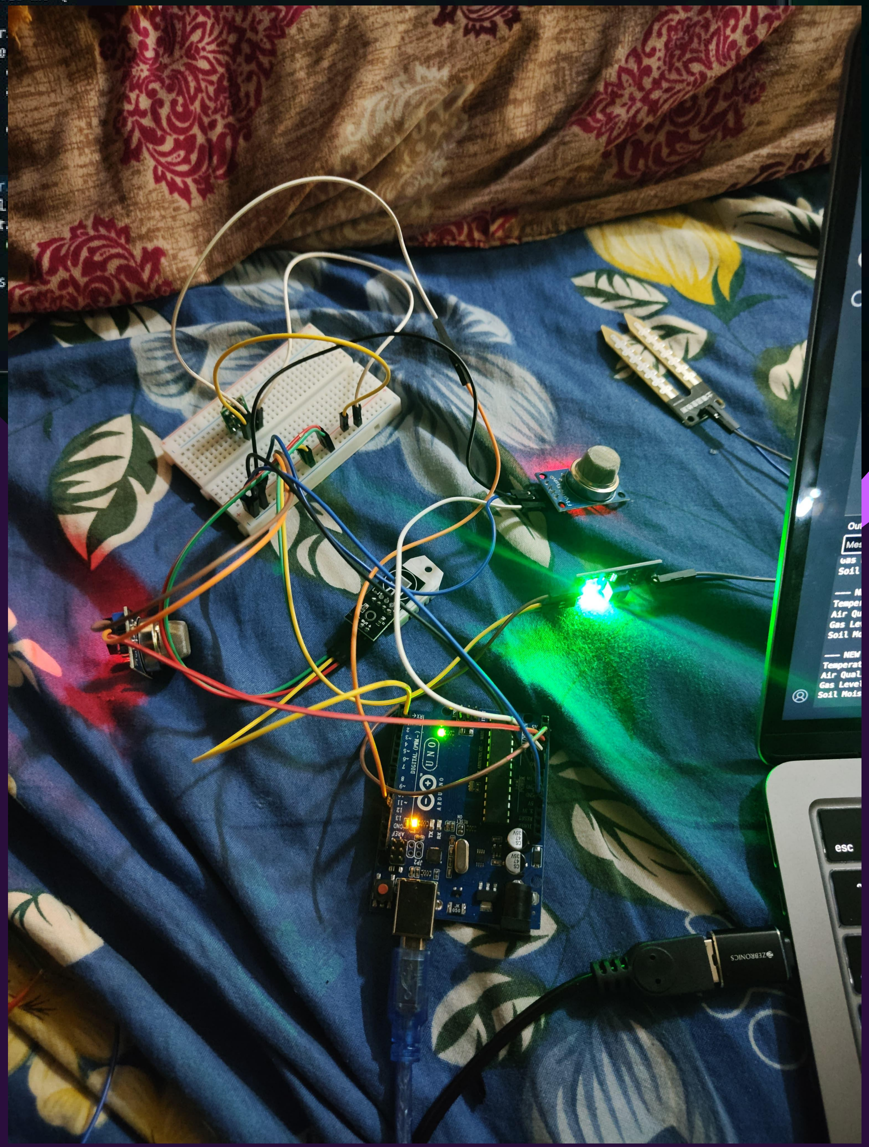
Completed



Circuit Assembly & Sensor Integration: Successfully assembled both Field Unit and Home Unit circuits on breadboards with all sensors (YL-69, DHT-22, MQ-5, MQ-135) properly connected to Arduino Uno boards.

RF Communication Setup: Established stable 433MHz wireless link between transmitter and receiver modules with successful data transmission tests.





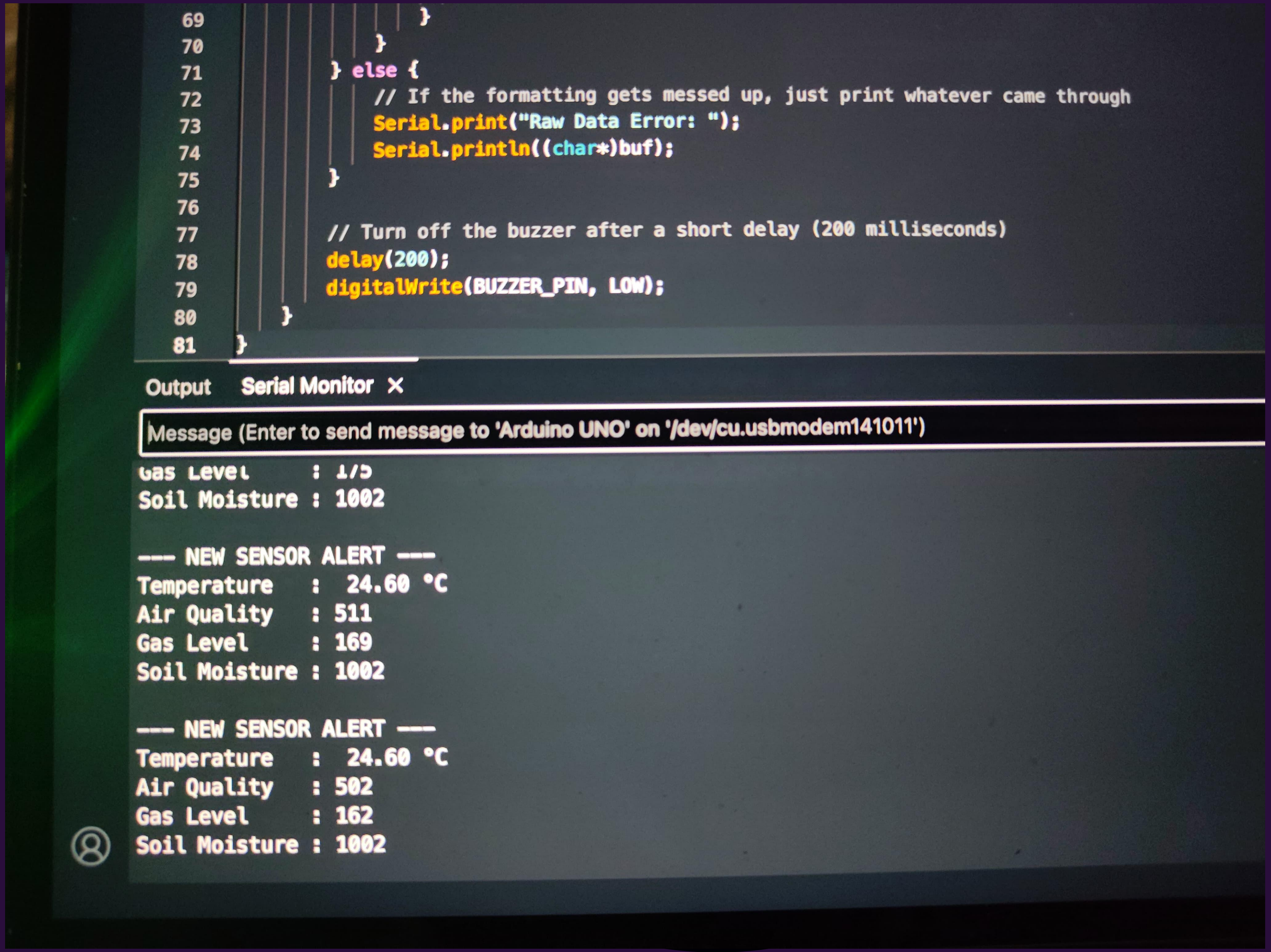
Code Implementation Screenshots

Transmitter Code

Key functions: dht.readTemperature(), analogRead(soilPin), rf_driver.send() for sensor data collection and RF transmission to home unit.

Receiver Code

Key functions: rf_driver.recv(), Serial.print() for displaying live data, digitalWrite(buzzerPin) for threshold alerts on dry soil or gas detection.



Results Screenshots & Live Output

Serial Monitor Display

Live sensor readings displayed on laptop via Arduino Serial Monitor showing real-time soil moisture %, temperature, humidity, and gas levels.

Alert Indicators

Buzzer activates when soil moisture drops below threshold or hazardous gas levels detected. Visual confirmation via Serial output timestamps.

Results
Analysis

RF Transmission Performance

Achieved reliable 433MHz wireless communication up to 100m range with consistent data packet delivery between field and home units.

Alert System Response

Buzzer triggers within 2 seconds of threshold breach; DHT-22 accuracy ±0.5°C; soil moisture readings calibrated to ±5% precision.

Work To Be Completed

Extended Range Testing

Test RF communication at various distances and obstacles to validate 100m+ range in real agricultural environments.

Enclosure & Power Design

Design weatherproof enclosures for field deployment and implement solar/battery power optimization for extended operation.

Field Deployment

Deploy complete system in actual farm conditions for real-world validation and long-term reliability testing.

Cost Analysis
& Budget

Electronics Components

Arduino Uno ×2 (₹800), 433MHz RF Tx/Rx (₹150), DHT-22 (₹250), YL-69 (₹100), MQ-135 (₹200), MQ-5 (₹180)

Accessories & Total

Buzzer (₹30), Breadboards ×2 (₹200), Jumper Wires (₹100). Total Estimated Budget: ₹2,010

Shopping List

Components Required

Microcontrollers & Communication

2x Arduino Uno R3, 433MHz RF Transmitter/Receiver Module Pair

Sensors & Alert Module

MQ-135 Air Quality, MQ-5 Gas/Fire, DHT-22 Temp/Humidity, YL-69 Soil Moisture, Piezo Buzzer

Supporting Components

Breadboards (2x), Jumper Wires (M-M, M-F), USB Cables for Arduino, Power Supply/Battery Pack

Project Conclusion

This RF-based Smart Agriculture Monitoring System delivers an affordable, offline solution for real-time farm monitoring without internet dependency.

Reliable 433MHz RF communication enables scalable sensor networks, empowering farmers with actionable environmental data.

Thank You

Questions & Discussion

We appreciate your attention throughout this presentation on the RF Based Smart Agriculture Monitoring System. We welcome any questions about our wireless sensor network, implementation approach, or future enhancements.

RF Smart Agriculture Team